Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Paul Mentink	**Email**	paul.mentink@tno.nl
Institute Name	TNO (Netherlands Organisation for Applied Scientific Research)			
Address	Automotive Campus 30, 5708 JZ Helmond, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Vehicle-in-the-Loop and Mixed Testing Platform for Heavy-Duty Off-Road and Defence Powertrains			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery & Materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Automotive & Railway Vehicles	Engines and powertrains	R&D & Engineering Services	Testing/Inspection/Analysis Techniques
Concept of Proposed Technology	TNO proposes a mixed virtual-physical validation platform approach that couples real powertrain hardware and test benches with real-time vehicle and environment simulation (vehicle-in-the-loop, ViL). The platform targets heavy-duty on-/off-road applications (construction, agriculture, trucks) and mission-critical defence mobility, where repeatable testing under extreme conditions is difficult, costly, or unsafe in the field.			

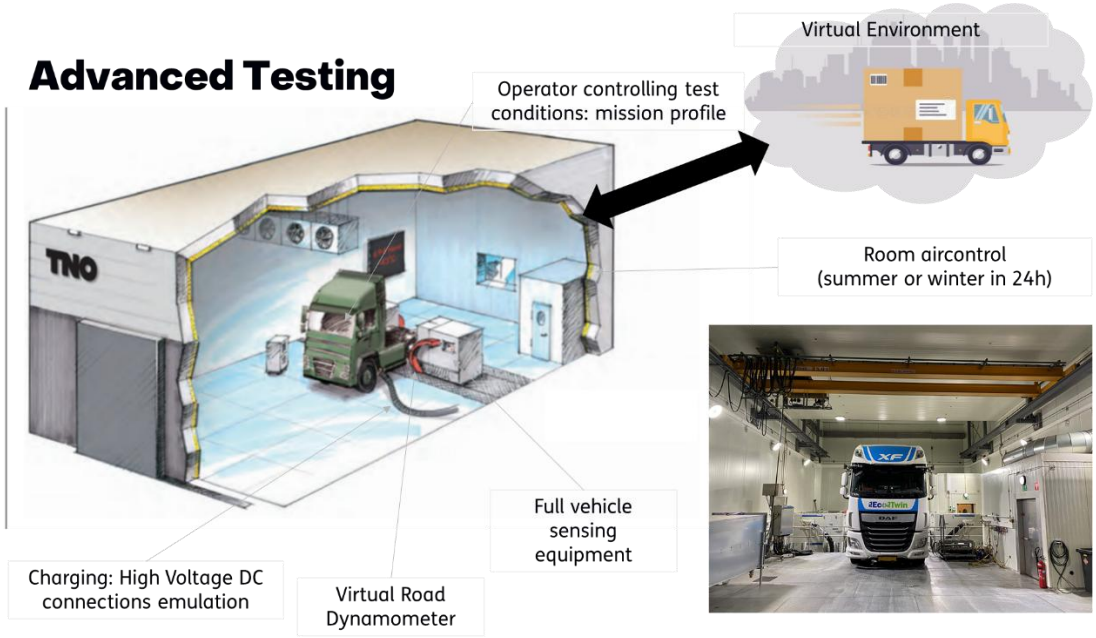

Figure 1 TNO coupling for physical and virtual testing of vehicles

Core concept: run a representative vehicle, powertrain or subsystem on hardware (engine, e-axle, battery pack, fuel cell module) while a real-time simulator computes driver demand, terrain loads, and environmental boundary conditions, closing the loop on torque, speed, thermal states, and energy flows.

Figure 2 TNO testing of various platforms

Key capabilities and differentiators at TNO (Helmond, ICSP and Battery Lab):

• Real-time co-simulation and control framework to connect chassis or powertrain dynamometers with validated vehicle and component models (TNO Powertrains modelling,

	control, and testing). [R7] • Climate and altitude heavy-vehicle testing: temperature -45 C to +55 C, air pressure down to 628 mbar (4000 m equivalent), and wind generation for load emulation. [R5] • Battery system testing up to pack and vehicle level with climate control and inert environments. Battery Lab supports hardware-in-the-loop testing and includes a 300 kW battery cycler. [R4] [R6] • ICSP capability for heavy-duty powertrains, including hydrogen and other renewable fuels, batteries and fuel cells for mobility use cases. [R3] • Defence, Safety and Security domain expertise for mission-critical validation and safety-by-design in defence mobility contexts – such as against NATO NG NRMM. [R8] [R9] The mixed testing approach supports Model-in-the-Loop, Software-in-the-Loop, Hardware-in-the-Loop, Powertrain-in-the-Loop and Vehicle-in-the-Loop workflows. It enables accelerated durability and performance validation by executing repeatable mission profiles while collecting synchronized high-frequency data for model calibration and certification-oriented evidence packages.
Rationale for Joint R&D Project	Heavy-duty off-road and defence vehicles face stringent requirements for reliability, safety, and operational availability. Electrification and alternative fuels add complexity: multi-energy architectures (battery, fuel cell, combustion, hybrids) require validation methods that cover electrical, thermal, control software, and environmental interactions. A joint R&D project is needed because it requires both high-fidelity methodology development (models, control, verification) and industrial-grade test equipment engineering and productization. TNO contributes model-based development and advanced validation methodology. This is demonstrated in peer-reviewed work by TNO researchers including Paul Mentink and Steven Wilkins on model-based co-design and energy management for electrified powertrains. [R2] TNO has also published model-based test methodology for distributed truck powertrains, including electrified trailers, in MTZextra. [R1] The Korean SME partner contributes domain-specific know-how and product platforms for heavy-duty chassis dynamometers and powertrain test systems, including applications in military and off-road sectors. This complements TNO facilities and methods, enabling a pathway from validated laboratory demonstrator to deployable test product and service.
Key Words	1. Vehicle-in-the-Loop (ViL) 2. Hardware-in-the-Loop (HiL)

	3. Heavy-duty off-road and defence mobility 4. Climate and altitude validation 5. Multi-energy powertrains (battery and hydrogen)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	R&D Objectives (technical): • Develop a real-time ViL integration layer (interfaces, timing, safety interlocks) to couple dynamometer control with vehicle and component simulation models. • Create a reference model library for heavy-duty off-road mission profiles, including terrain load models, towing and payload, and duty cycles relevant to non-road and defence operations. • Integrate environmental boundary conditions (temperature, pressure, wind) with the test control loop to run combined climate-altitude and mission-profile validation. [R5] • Enable multi-energy configurations in the loop: battery pack and fuel cell modules integrated with powertrain control, using HiL-capable laboratory equipment and a 300 kW battery cycler where relevant. [R4] [R6] • Define a verification and validation protocol producing repeatable evidence for performance, efficiency, durability, and safety, aligned with certification-oriented testing at ICSP. [R3]			
R&D Target Specifications	R&D Target Specifications (initial): • Closed-loop real-time operation: loop latency target below 50 ms for torque and speed commands between simulator and test bench (to be confirmed during integration). • Heavy-vehicle climate-altitude capability: temperature -45 C to +55 C; pressure down to 628 mbar (4000 m equivalent); wind generation 120 km/h at 1 m2. [R5] • Electrified powertrain capability: up to 750 V DC and 400 A with 300 kW bidirectional charge/discharge for pack and vehicle testing. [R5] • Battery Lab environment: climate-controlled testing spaces up to pack and EV level, temperature -40 C to +150 C, with inert test environments and gas detection; 300 kW battery cycler; HiL capable. [R4] [R6] • Model accuracy targets: key performance metrics within 10% between simulation and			

	measured hardware results over representative mission profiles after calibration.
R&D Plan	R&D Plan: • **Phase 1 (Months 1-6): Requirements and architecture.** Define mission profiles and interfaces. Establish real-time co-simulation framework and safety concept. Deliverable: ViL architecture specification and initial model set. • **Phase 2 (Months 7-18): Bench integration.** Connect SME dynamometer and control systems with TNO simulation environment and DAQ. Implement timing, synchronization, and fail-safe mechanisms. Deliverable: integrated ViL prototype in lab. • **Phase 3 (Months 19-30): Mixed testing campaigns.** Execute climate-altitude mission tests with representative heavy-duty powertrain configurations (including electrified and hydrogen-relevant cases). Calibrate models using test data. Deliverable: validated mixed testing methodology and datasets. • **Phase 4 (Months 31-36): Demonstration and transfer.** Demonstrate repeatability and performance against targets. Produce engineering package and deployment guidelines for productization and future certification use cases. Deliverable: TRL 6 demonstrator and exploitation plan.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Expected contribution from TNO: • Technical leadership for ViL methodology, including modelling, control design, and validation protocol definition. [R7] • Provide access to ICSP and Battery Lab facilities, including climate-altitude heavy-vehicle testing and HiL-capable battery testing infrastructure. [R3] [R4] [R5] [R6] • Develop and maintain the real-time co-simulation environment and digital twin models used for mission profile execution and calibration. • Execute test campaigns, analyze results, and produce evidence packages suitable for certification-oriented discussions. • Support defence-relevant safety, security, and reliability considerations via TNO Defence, Safety and Security expertise. [R8]
Expected R&R of Korean SME Partner	Expected contribution from Korean SME partner: • Provide industrial-grade chassis dynamometer and/or powertrain test system platforms and engineering resources for integration with ViL control and data interfaces. • Implement and validate real-time control interfaces (torque, speed, road load emulation) and ensure equipment safety interlocks meet high-power and mission-critical requirements. • Contribute heavy-duty off-road and defence-oriented test scenarios and acceptance criteria based on market needs and end-user requirements. • Co-develop a productization pathway for the mixed testing platform enabling commercialization after TRL 6 demonstration.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes and impact: • TRL 6 vehicle-in-the-loop and mixed testing demonstrator for heavy-duty off-road and defence powertrains, validated in a relevant environment (lab-based climate-altitude and high-power electrical testing). [R5] [R6] • Repeatable validation methodology and reference mission-profile library for accelerated development and evidence generation for heavy-duty electrified and hydrogen powertrains. • Reduced risk prior to field trials by shifting early verification to controlled, repeatable lab scenarios, supporting faster iteration on hardware and control software. [R7] • Commercialization pathway for the Korean SME partner to offer ViL-enabled dynamometer solutions for domestic and international heavy-duty markets, including defence and non-road mobile machinery. • Strengthened Korea-Netherlands collaboration in advanced mobility and energy transition, with potential spillover to standards and certification practices.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization